

SECTION B

7 (short)

(a) A composite material has a matrix of polypropylene and contains 10% glass reinforcement by volume. Polypropylene has a Young's modulus of 1 GPa, while glass has a Young's modulus of 70 GPa.

(i) Estimate the Young's modulus of the composite along the fibre direction, assuming the glass reinforcement is in the form of long, parallel fibres. [3]

(ii) Estimate the Young's modulus of the composite, assuming the glass reinforcement is in the form of small dispersed particles. [3]

(b) A cubic element of a linear elastic material with Young's modulus E and Poisson's ratio ν is subjected to uniaxial tensile stresses $\sigma_1 = \sigma$, $\sigma_2 = 0$ and $\sigma_3 = 0$. Find an expression of volumetric strain (or dilation) Δ . For what value of Poisson's ratio is the volume of the element conserved? [4]